

# 3.2 Hydrocarbons

## Question Paper

|            |                      |
|------------|----------------------|
| Course     | CIEA Level Chemistry |
| Section    | 3. Organic Chemistry |
| Topic      | 3.2 Hydrocarbons     |
| Difficulty | Hard                 |

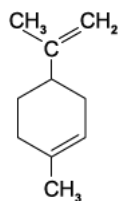
**Time allowed:** 20

**Score:** /10

**Percentage:** /100

## Question 1

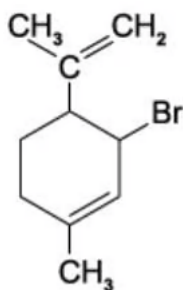
Limonene is an oil formed in the peel of citrus fruits.



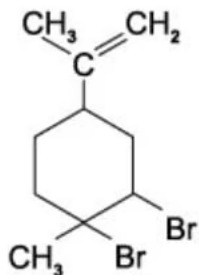
Limonene

Which product is formed when an excess of bromine,  $\text{Br}_2(l)$ , reacts with limonene at room temperature in the dark?

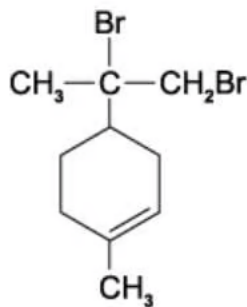
A.



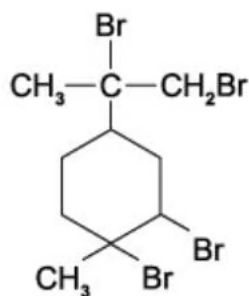
B.



C.



D.

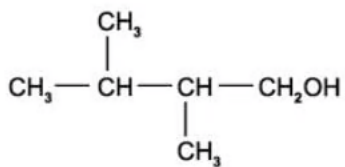


[1 mark]

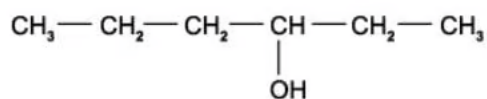
## Question 2

Which isomer of  $C_6H_{13}OH$  gives the greatest number of different alkenes when it is dehydrated?

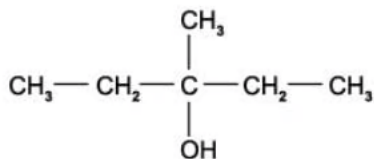
A.



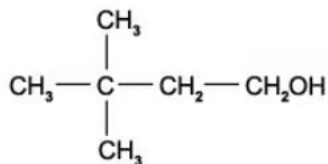
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C.



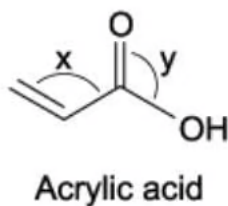
D.



[1 mark]

### Question 3

Acrylic acid is produced from propene, a gaseous product of oil refineries.



Which statement about acrylic acid is **not** correct?

- A. Both bond angles x and y are approximately  $120^\circ$ .
- B. It decolourises aqueous bromine.
- C. It gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent.
- D. It reacts with an alcohol to give an ester.

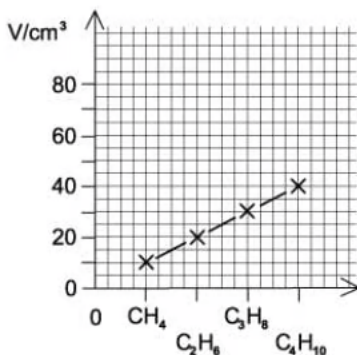
[1 mark]

## Question 4

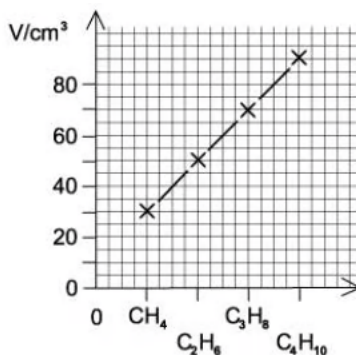
Samples of  $10\text{ cm}^3$  of each of the first four members of the alkane series are separately mixed with  $70\text{ cm}^3$  of oxygen. Each is then burned and the total volume,  $V$ , of residual gas measured again at room temperature and pressure.

Which graph represents the results that would be obtained?

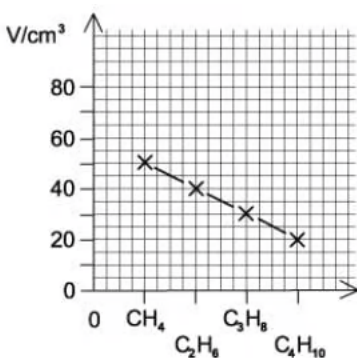
A.



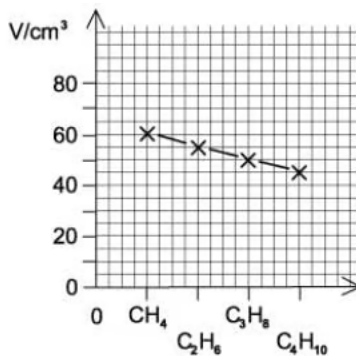
B.



C.



D.



[1 mark]

### Question 5

Use of the Data Booklet is relevant to this question.

Which compound has an  $M_r$  of 84 and will react with HBr to give a product with an  $M_r$  of 164.9?

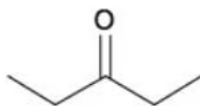
A.



B.



C.



D.



[1 mark]

### Question 6

Alkane **X** has molecular formula  $C_4H_{10}$ .

**X** reacts with  $Cl_2$  (g) in the presence of sunlight to produce only two different monochloroalkanes,  $C_4H_9Cl$ . Both of these monochloroalkanes produce the same alkene **Y**, and no other organic products, when they are treated with hot ethanolic KOH.

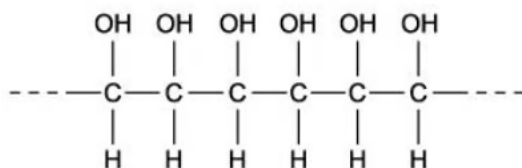
What is produced when **Y** is treated with hot concentrated acidified  $KMnO_4$ ?

- A.  $CO_2$  and  $CH_3CH_2CO_2H$
- B.  $CO_2$  and  $CH_3COCH_3$
- C.  $HCO_2H$  and  $CH_3COCH_3$
- D.  $CH_3CO_2H$  only

[1 mark]

### Question 7

The following diagram represents the structure of a possible polymer.



By which method might this polymer be made?

- A. Polymerise ethene followed by hydration.
- B. Polymerise ethene followed by oxidation with cold acidified  $KMnO_4$ .
- C. Polymerise 1,2-dichloroethene followed by hydrolysis.
- D. Polymerise 1,2-dichloroethene followed by oxidation with cold acidified  $KMnO_4$ .

[1 mark]

### Question 8

The petrol additive tetraethyl-lead (IV),  $\text{Pb}(\text{C}_2\text{H}_5)_4$ , is now banned in many countries. When it is completely burned in air, lead (II) oxide,  $\text{CO}_2$  and  $\text{H}_2\text{O}$  are formed.

How many moles of oxygen are required to burn one mole of  $\text{Pb}(\text{C}_2\text{H}_5)_4$ ?

- A. 9.5
- B. 11
- C. 13.5
- D. 27

[1 mark]

### Question 9

Oxidation of an alkene **Y** gives a diol; further oxidation gives a diketone.

What could be **Y**?

- A.  $\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)_2$
- B.  $(\text{CH}_3)_2\text{CHCH}=\text{CH}_2$
- C.  $\text{C}_6\text{H}_5\text{CH}=\text{CHC}_6\text{H}_5$
- D.  $(\text{C}_6\text{H}_5)_2\text{C}=\text{CHCH}_3$

[1 mark]

### Question 10

In a preparation of ethene, ethanol is added a drop at a time to a heated reagent **Y**. To purify the ethene it is bubbled through a solution **Z** and then collected.

What could reagent **Y** and solution **Z** be?

|   | Reagent Y                                   | Solution Z                           |
|---|---------------------------------------------|--------------------------------------|
| A | Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ | Dilute NaOH                          |
| B | Concentrated $\text{H}_2\text{SO}_4$        | Dilute $\text{H}_2\text{SO}_4$       |
| C | Concentrated $\text{H}_2\text{SO}_4$        | Dilute NaOH                          |
| D | Ethanollic NaOH                             | Concentrated $\text{H}_2\text{SO}_4$ |

[1 mark]

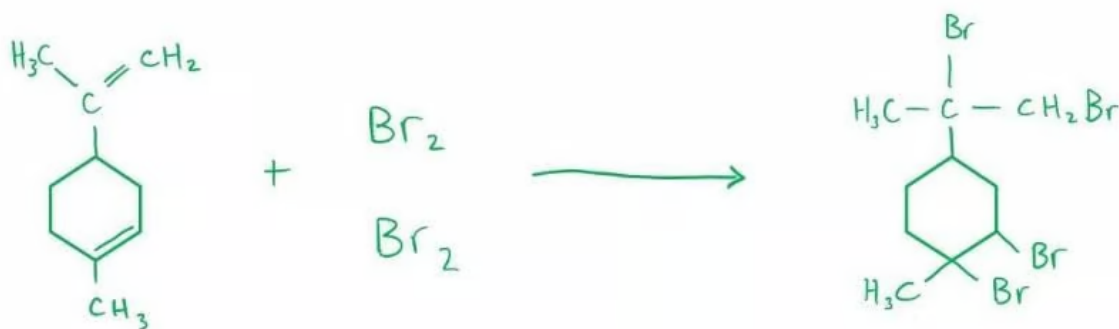


## Model Answers: Hard

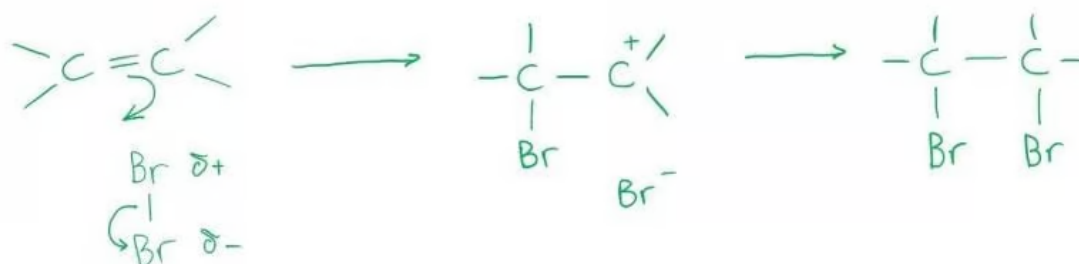
1

The correct answer is **D** because:

- Limonene is an alkene with two double bonds in its structure.
- When reacted in an excess of bromine,  $\text{Br}_2$ , electrophilic addition occurs between the bromine and the double bonds.



Electrophilic addition mechanism:

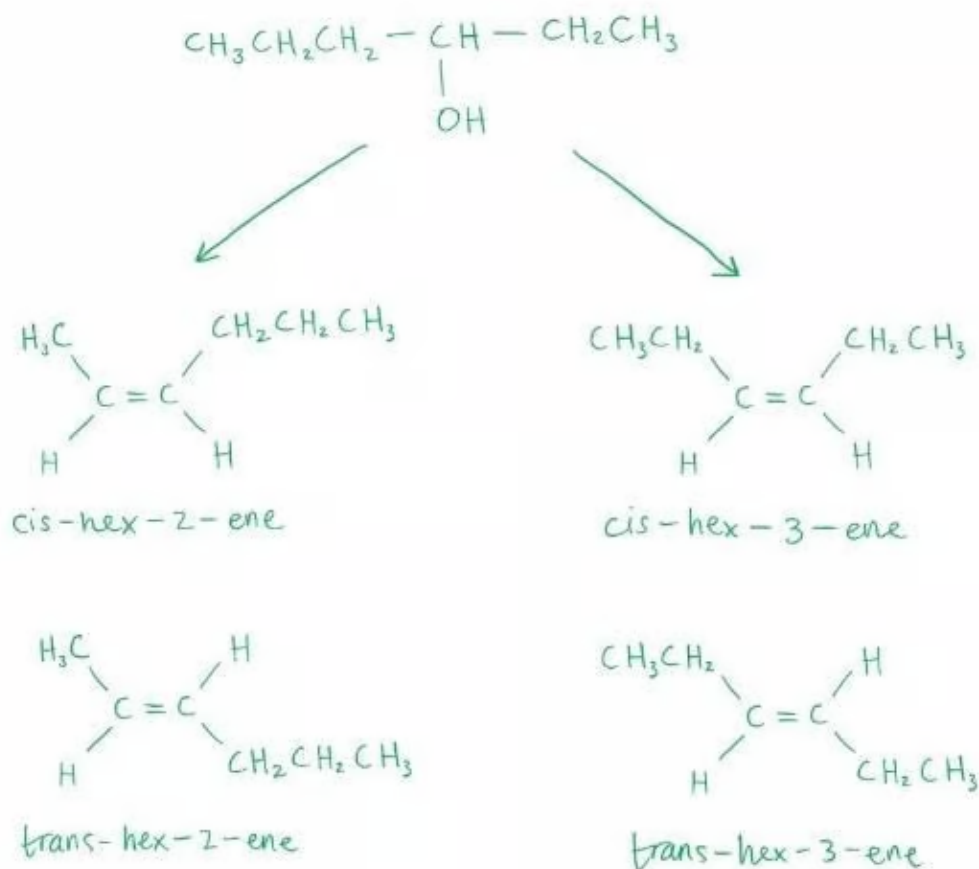


2

The correct answer is **B** because:

- Alcohols can be dehydrated to form alkenes by losing a water molecule to form a double bond.

- Consider option **B**:
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$  will give you 2 isomers:
  - 2-hexene and 3-hexene
- PLUS, the stereoisomers of these:
  - Cis-hex-2-ene and trans-hex-2-ene
  - Cis-hex-3-ene and trans-hex-3-ene
- Therefore, option **B** gives the greatest number of different alkenes.



**A & D** is incorrect as these are primary alcohols, so they will produce less isomers.

**C** is incorrect as  $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})\text{CH}_2\text{CH}_3$  will give you 2 isomers 2-methylpent-2-ene and 2-ethylbut-1-ene and there are two ways you can produce 2-methylpent-2-ene, but this is still fewer isomers than option **B**.

The correct answer is **C** because:

- 2,4-dinitrophenylhydrazine (Brady's reagent) is a test for the carbon-oxygen double bond found in aldehydes and ketones.
- Although acrylic acid is a carboxylic acid and contains the C=O bond, Brady's reagent only works on aldehydes and ketones.
- Therefore, it would not produce an orange precipitate.

**A** is incorrect as each carbon in acrylic acid contains 3 bonding groups and no lone pairs of electrons, so, both bond angles, x and y, are equal to approximately 120°.

**B** is incorrect as acrylic acid contains one C=C double bond, which opens up in the presence of aqueous bromine and hence decolourises it.

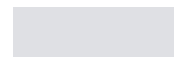
**D** is incorrect as acrylic acid is a carboxylic acid, which reacts with an alcohol to give an ester.

4

The correct answer is **D** because:

- The first four members of the alkane series are
  - Methane (CH<sub>4</sub>), ethane (C<sub>2</sub>H<sub>6</sub>), propane (C<sub>3</sub>H<sub>8</sub>) and butane (C<sub>4</sub>H<sub>10</sub>)
- The combustion equation for methane is as follows:
  - $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- We have 10 cm<sup>3</sup> of CH<sub>4</sub> and 70 cm<sup>3</sup> of O<sub>2</sub>
  - So, 10 cm<sup>3</sup> of CH<sub>4</sub> reacts with 20 cm<sup>3</sup> of O<sub>2</sub>
- This leaves 50 cm<sup>3</sup> of O<sub>2</sub> leftover and 10 cm<sup>3</sup> of CO<sub>2</sub> will be produced.
- The residual volume of gas remaining at the end of the reaction is therefore:
  - 50 + 10 = 60 cm<sup>3</sup>
- The combustion equation for ethane is as follows:
  - $\text{C}_2\text{H}_6 + \frac{7}{2}\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
- We have 10 cm<sup>3</sup> of CH<sub>4</sub> and 70 cm<sup>3</sup> of O<sub>2</sub>

- So,  $10\text{ cm}^3$  of  $\text{CH}_4$  reacts with  $35\text{ cm}^3$  of  $\text{O}_2$
- This leaves  $35\text{ cm}^3$  of  $\text{O}_2$  leftover and  $20\text{ cm}^3$  of  $\text{CO}_2$  will be produced.
- The residual volume of gas remaining at the end of the reaction is therefore:
  - $20 + 35 = 55\text{ cm}^3$
- Therefore, graph **D** is correct as we can see the first two points are  $60\text{ cm}^3$  and  $55\text{ cm}^3$

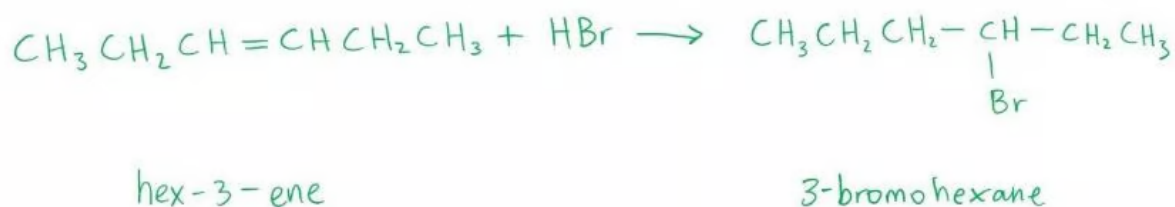


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5

The correct answer is **D** because:

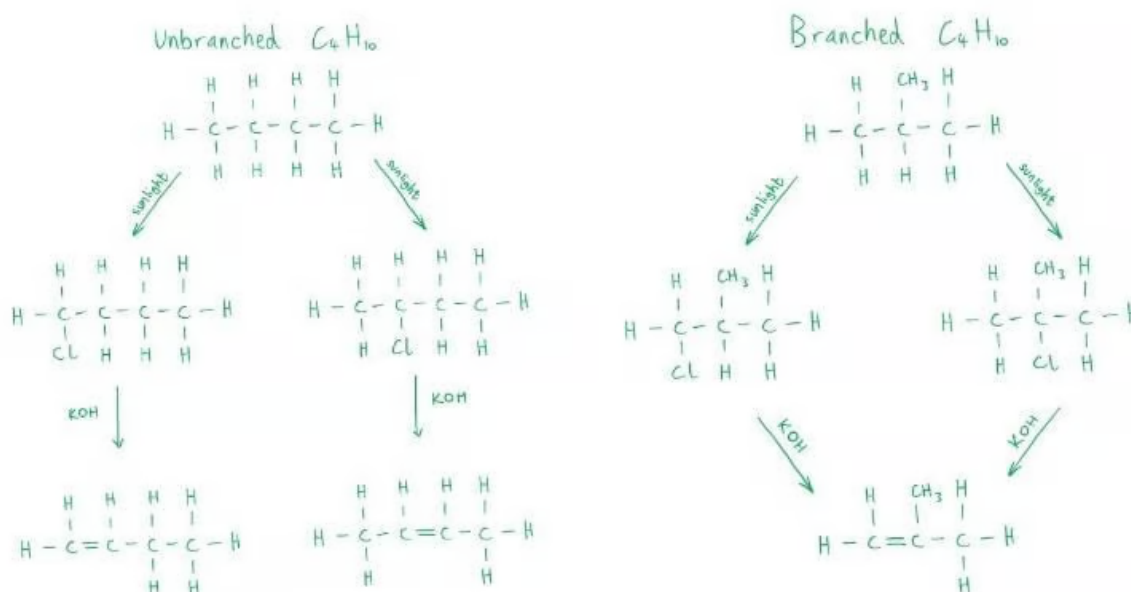
- Bromine is an electrophile (electron loving species).
- It reacts with alkenes via electrophilic addition by breaking the C=C double bond.
- Therefore, HBr will only react with compound **B** or **D**.
- Consider compound **D** (hex-3-ene):
  - $M_r$  of 6 carbons =  $6 \times 12 = 72$
  - $M_r$  of 12 hydrogens =  $12 \times 1 = 12$
- So,  $M_r$  of hex-3-ene =  $72 + 12 = 84$
- When hex-3-ene reacts with HBr it forms 3-bromohexane.
- The  $M_r$  of 3-bromohexane is:
  - $M_r$  of 6 carbons =  $6 \times 12 = 72$
  - $M_r$  of 13 hydrogens =  $13 \times 1 = 13$
  - $M_r$  of 1 bromine = 79.9
- So,  $M_r$  of 3-bromohexane =  $72 + 13 + 79.9 = 164.9$ 
  - Therefore, the correct compound is **D**.



6

The correct answer is **B** because:

- Alkane **X** must be a branched compound in order for the same alkene to be produced from two different monochloroalkanes.
- If alkane **X** were not branched, there's a possibility two alkenes would be produced:
  - So, alkane **X** is 2-methylpropane.
- It undergoes free radical substitution with  $\text{Cl}_2$  in sunlight to produce 1-chloro-2-methylpropane and 2-chloro-2-methylpropane.
- Both of these compounds, after undergoing dehydrohalogenation, produce the same alkene, 2-methylprop-1-ene.
- The diagram shows an example of an unbranched alkane **X** forming two possible monochloroalkanes and then two different alkenes.
- Two different alkenes would also be produced if the 2nd and 3rd hydrogen atoms were substituted by chlorine instead of the 1st and 2nd.
  - So, the compound must be branched to obtain the same alkene from the monochloroalkanes:



- Alkene **Y**, 2-methylpropene, then undergoes vigorous oxidation with hot concentrated acidified  $\text{KMnO}_4$  to produce  $\text{CO}_2$  and  $\text{CH}_3\text{COCH}_3$ .
  - Therefore, the answer is **B**.

7

The correct answer is **C** because:

| Step                                                                                   | Working out                                              |
|----------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1. Draw the monomer structural formula                                                 | <p>1,2-dichloroethene</p>                                |
| 2. Break the double bond between the carbon atoms to form the base unit of the polymer |                                                          |
| 3. Replace the -C/with -OH                                                             | <p>hydrolysis using NaOH(aq) and heated under reflux</p> |

**A** is incorrect as once we break open the C=C bond to form poly(ethene), hydrating the polymer won't add -OH groups to the chain.

**B & D** is incorrect as oxidation with cold acidified  $\text{KMnO}_4$  is the method to change alkenes into diols, it wouldn't work with either polymerised ethene or 1,2-dichloroethene.

8

The correct answer is **C** because:

- The general equation for complete combustion is:
  - $C_xH_y + O_2 \rightarrow CO_2 + H_2O$
- So, the equation for complete combustion of  $Pb(C_2H_5)_4$  is:
  - $Pb(C_2H_5)_4 + O_2 \rightarrow PbO + CO_2 + H_2O$

The correct answer is **C** because:

| Step                                                                                   | Working out                                              |
|----------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1. Draw the monomer structural formula                                                 | <p>1,2-dichloroethene</p>                                |
| 2. Break the double bond between the carbon atoms to form the base unit of the polymer |                                                          |
| 3. Replace the -C/with -OH                                                             | <p>hydrolysis using NaOH(aq) and heated under reflux</p> |

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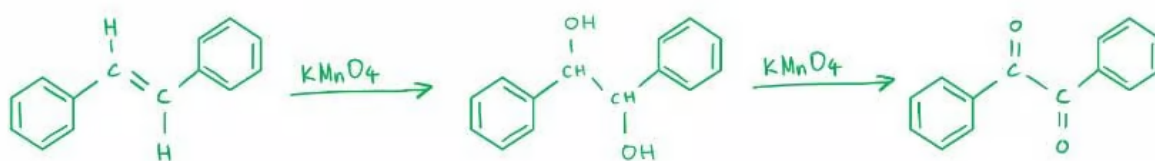
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- So, the equation for complete combustion of  $Pb(C_2H_5)_4$  is:
  - $Pb(C_2H_5)_4 + O_2 \rightarrow PbO + CO_2 + H_2O$
- Balancing the equation for one mole of  $Pb(C_2H_5)_4$ :
  - $Pb(C_2H_5)_4 + \frac{27}{2}O_2 \rightarrow PbO + 8CO_2 + 10H_2O$
- Therefore, 13.5 moles of oxygen are required to burn one mole of  $Pb(C_2H_5)_4$

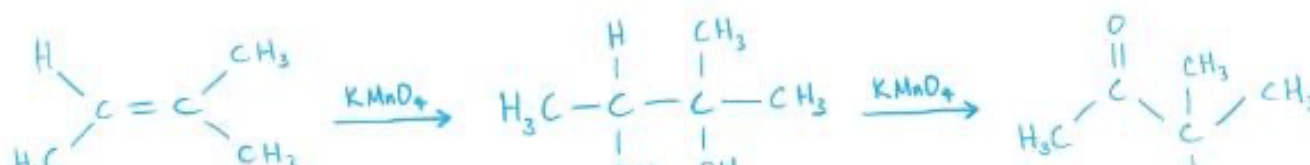
9

The correct answer is **C** because:

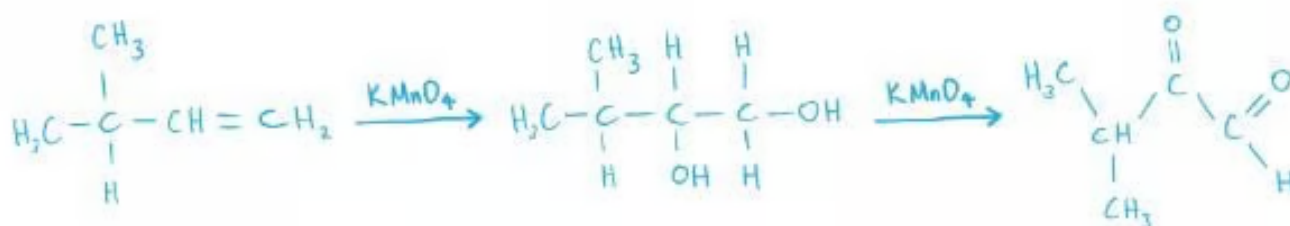
- To form a diol, the alkene would require that the unsaturated carbons each have at least 1 hydrogen atom bonded to it.
- Option **C** provides a secondary alcohol on both sides hence will produce a diketone, making it the correct option.



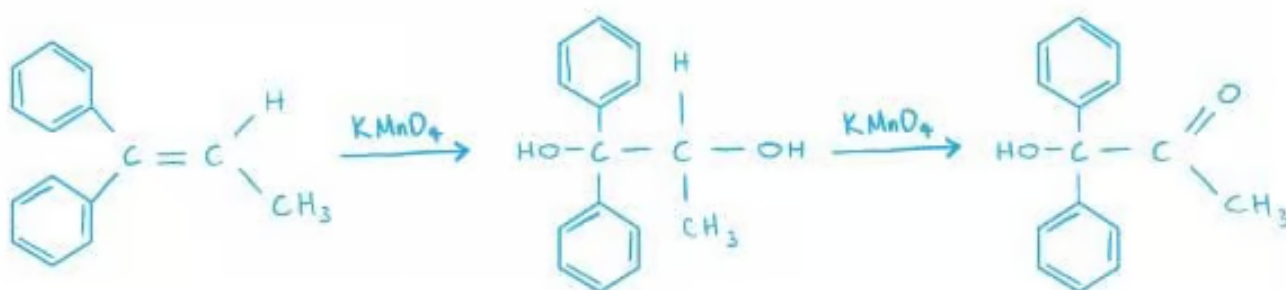
**A** is incorrect as the first oxidation opens the double bond and two hydroxyl groups ( $-OH$ ) form. The one on the right is attached to a carbon that has no hydrogen atoms and so cannot be further oxidised.



**B** is incorrect as the first oxidation opens the double bond and two hydroxyl groups (-OH) form. The one on the right is at the end of the chain and further oxidation would produce an aldehyde.



**D** is incorrect as the first oxidation opens the double bond and d two hydroxyl groups (-OH) form. The -OH group that is attached to the LHS is on a carbon that has no hydrogen atoms, so, it cannot undergo further oxidation.



10

The correct answer is **C** because:

- Alcohol undergoes dehydration to form an alkene.
- Concentrated  $\text{H}_2\text{SO}_4$  is among the possible conditions for this reaction.
  - This eliminates options **A** and **D** as possible answers.
- After dehydration occurs there may be remaining  $\text{H}_2\text{SO}_4$  mixed with the alkene

which must be neutralised.

- $\text{H}_2\text{SO}_4$  is acidic and in option **B**, the addition of more  $\text{H}_2\text{SO}_4$  will contaminate the alkene more, so, **B is incorrect**.
- A basic solution will neutralise the acid, such as the dilute  $\text{NaOH}$  in option **C**, which is the correct answer.

